

# Qinghua Sun

San Diego, CA | qis010@ucsd.edu | +1 747-222-0517 | treehouserly.asia | github.com/QinghuaSun1127

## EDUCATION

### University of California San Diego

M.S., Electrical and Computer Engineering (Signal and Image Processing); GPA: 4.00/4.00

Expected Dec 2027

San Diego, CA

### University of Electronic Science and Technology of China

B.Eng., Communication Engineering; Glasgow College Joint Program; GPA: 3.56/4.00

Jun 2025

Chengdu, China

## TECHNICAL SKILLS

**Languages:** Python, TypeScript/JavaScript, Go, SQL, C/C++, MATLAB

**AI / ML:** PyTorch, LangGraph, RAG, LLM evaluation, LoRA fine-tuning, streaming ASR

**Software / Tools:** FastAPI, React, Next.js, PostgreSQL/pgvector, Redis, Docker, Git, Linux, WebRTC, pytest

## PROFESSIONAL EXPERIENCE

### AIBuildAI (UC San Diego spinout)

Research Contributor

Jul – Sep 2026

San Diego, CA

- Evaluated AIBuildAI Max for co-founder Prof. Pengtao Xie on a lab NVIDIA A100; diagnosed environment, stage-timeout, and output-alignment failures until the autonomous setup-to-aggregation pipeline completed.
- Verified 16,500 CUDA training steps, hidden-holdout evaluation, and a 5,000-row submission with exact ID alignment on a synthetic test; authored a world-model survey and benchmark proposal.

### ChainAI

AI Agent and Real-Time Systems Intern

Jun – Sep 2026

- Built a Python/Chrome DevTools Protocol test harness for a trading voice agent; automated 94 multi-account turns and captured ASR transcripts, audio, screenshots, browser logs, and latency.
- Curated 5,604 facts from 60 mock accounts and evaluated 60 agent cases with 180 LLM-judge votes; manually reviewed tool traces, factual errors, and privacy failures.
- Contributed to a four-GPU streaming ASR release; in targeted tests, long-utterance error fell from 26.32% to 3.16% and BTCUSDT recognition error from 27.78% to 0%; verified four concurrent streams.
- Prototyped adaptive media delivery in Go/Pion WebRTC using GCC feedback and active probing; validated 50-image transfer with per-image acknowledgments.

### Shanghai Tenglong Technology Co., Ltd. (Anna)

App Development Intern

May – Jul 2026

Shanghai, China

- Built AnnaWrite in React/TypeScript with AI drafting, sentence-level rewrites, version restore, and DOCX/PDF export; integrated `llm.complete` and validated a staging build in the browser.
- Built AnnaVisual's web UI and Python workflow to turn brand references into editable profiles and image briefs while preserving fixed logo and mascot assets.
- Implemented seven props-driven chart/table widgets and agent-callable contracts for presentation templates; verified that generated PPTX charts remained editable.

### Shanghai Yunwu Intelligent Technology Co., Ltd. (Feeling AI)

AI Application and Algorithm Intern

Oct 2025 – Mar 2026

Shanghai, China

- Profiled and optimized Flimo's 3D virtual-human text/speech/motion pipeline, reducing end-to-end latency by 20% in internal tests.
- Prepared instruction-tuning data and LoRA fine-tuning workflows; stress-tested long multimodal sessions and documented 50+ audio-visual timing failures.

## SELECTED PROJECTS

### AgentFlow Enterprise AI Agent Platform

Independent Developer | [github.com/QinghuaSun1127/agentflow-platform](https://github.com/QinghuaSun1127/agentflow-platform)

Mar – Apr 2026

- Built a LangGraph agent platform with intent routing, pgvector-backed document RAG, Redis memory, FastAPI endpoints, and a Streamlit chat UI.
- Added JWT authentication, role-based access, DLP filtering, rate limiting, and health/metrics endpoints; added Docker deployment and pytest/CI checks.

### “Tree Hole” Multimodal AI Emotional Companion

Independent Developer | [Live demo](#) | [Source code](#)

Feb – Apr 2026

- Implemented encrypted local-first memory (PBKDF2 + AES-GCM), erasure controls, and an emotion-responsive Pixi.js/WebGL interface; deployed a live demo.

**Honors:** UESTC Excellent Student Cadre; Hainan Province Entrepreneurship and Innovation Competition, Second Class Prize.